MILIRARY SPECIFICATION

MOUNT, HOWITZER, M182

This specification is approved for use within the U..S. Army Armament, Munitions and Chemical Command and is available,.. for use by all Departments and Agencies of the Department of Defense.

1. SCOPE .

1.1 Scope. This specification covers one type of hydrop pneumatic gun mount for use in combat vehicles such as the M109 such as the M109 SPH .

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Departrnent of Defense Index of Specifications and Standards (DODISS) and sUPPlement thereto, cited in the solicitation (see 6.2) .

SPECIFICATIONS

FEDERAL

BB-N-441	Nitrogen, Technical

MILITARY

MIL-H-6083	Hydraulic Fluid, Petroleum Base Preservation and Testing
MIL-G--10924	Grease, Automotive and Artillery
MIL-w-63150	Weapons and Support Materiel, Standard Quality Assurance Provisions For
MIL-A-70625	Automated Acceptance Inspection Equipment Design, Testing and Approval of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be on use in improving this document should be addressed to: Commander,U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA. Picatinny Arsenal, New Jcscy 07806-5000 by using the self-addrressed Standardization Document improvement Proposal (DDForm1426)appearing at the end of this document or by letter

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STANDARDS

MILITARY

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-130	Identification Marking of U.S. Military Property

(Unless otherwise indicated, copies of the above specifications and standards are available from the Naval Publications and Forms Center, (Attn: NPODS). 5801 Tabur Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DOCUMENTS

MWO 9-2350-311-55-4	Modification, Gun Mount, M178 to M182

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12910860	Modification kit, M182 Gun Mount

PUBLICATIONS

P12910860	Packaging data sheet for Mount, Gun, M182

(Unless otherwise indicated, copies of the above documents, drawings, and publications are aavailable from the U.S. Army ARDEC, Attn: SMCAR-ESC-AA, Picatinny Arsenal, NJ 07806-5000.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issue's of "the-documents which

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are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTX)

ASTM D1744 - Petroleum Products, Liquid, Water In. By Karl Fisher Regent

(Application for copies should be addressed to the American society for Testing-and Materials, 1916 Race Street, Philadelphia, PA 19103-1137.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documeents also may be available in or through libraries or other informational services.)

2.3 order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable lasws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS .

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.2).

3.2 Materials. Materials shall conform to the material specifications referenced on the applicable drawings. Certification of material may be accepted in lieu of material tests. The contractor shall be responsible for determining that materials assemblies, and parts requiring product qualifica-tions are listed or qualified for inclusion on the applicable Qualified Products Lists (QpL) l In the instance that a qualified vendor no longer exists on the QPL, the contractor is responsible for selecting a vendor whose product meets the minimum requirement of the specification. The contractor must also submit to the Government for approval documentation which confirms the vendors ability to meet the specification in question. Contractor's inspection record shall specifically list such components, name of supplier (s) , and number and date of applicable QPL or Government approved vendors not on a QPL.

3.3 Construction. The M182 Gun Mount, and all of its assemblies and components, shall be constructed as specified in MWO 9-2350-311-55-4 with dimensions and tolerances as specified on the applicable drawings referenced on Drawing 12910860.

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3.3.1 Hydraulic system.

3.3.1.1 Particle size. The recoil mechanism shall be filled with hydraulic fluid, bled, and refilled until all air and measurable moisture are eliminated. The initial and subsequent filling of the hydraulic system shall be with hydraulic oil conforming to MIL-H-6083 and filtered through a 5 micron filter. Immediately after mechanical gymnastication, not to exceed 5 minutes, a 100 ml sample of hydraulic oil shall be taken from the lowest drain hole of each recoil cylinder. Samples shall be drained into a suitable container. Each sample must meet the requirements of Table I.

TABLE I. Hydraulic Fluid cleanliness.

The water content of the hydraulic fluid shall not be more than 0.1% as determined by standard method of test IAW ASTM D1744. The hydraulic fluid shall be free of fforeign materials within the following limits:

Particle Size	Quantity Allowed
below 50 microns	no limit
50 - 100 microns	154
101 - 200 microns	16
201 - 400 microns	2

Particles over 400 microns are not acceptable unless they are fiber or lint. Only two particles of fiber or int are acceptable, provided they do not exceed the dimensions of 40 microns in width and 3000 microns in length.

L3.1.2 Leakage. During cyclying and after all cycling, no indication of unusual hydraulic leakage shall occur.

3.3.2 Surface cleanliness. All finish machined surfaces shall be free of condensation or other foreign residues from the time of final machining until final assembly. All parts and assemblies being assembled into the completed mount shall be free of all foreign material after assembly. Assembly shall be performed in the controlled environment specified in 6.3.

3.3.3 Part modification. No parts shall be altered or hand-fitted during assembly unless such modification has been authorized by an approved drawing change, or is authorized by instructions presently listed on the drawings.

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3 .3.4 Air, gas and accumulator system. The gas system shall be clean and free of all foreign matter. The water content of the air used to blow the lines shall not exceed *one* percent of the total volume of the air used. The system shall be filled, bled, and refilled until all air and measurable moisture are eliminated.

3.3.5 Cylinder pressure. The cylinders of the mount shall withstand the pressures specified on the drawings without damage, permanent deformation or leakage.

3.3.6 Lubrication. Lubricants conforming to MIL-G-10924 shall be used for the assembled mount.

3.3.7 Gas. The nitrogen used to charge and fill the gun mount recuperator and replenisher shall conform to BB-N-441, Type I, Grade B, Class I.

3.4. Performance.

3.4.1 Mechanical gymnastication. Each mount shall be subjected to the mechanical gymnastication test as specified in 4.6.1 without evidence of erratic movement and shall meet the requirements of 3.4.1.1 and 3.4.1.2. During and after gymnastication, the gun mount shall meet the requirement of 3.3.1, 3.4.5 and 3.4.6.

3.4.1.1 Counter recoil (CTRR) time. The CTRR time for each mechanical gymnastication shall not exceed the requirements specified below.

Elev. in Degree ,	CTRR stroke Time (Seconds)	Buffer Stroke Time (Seconds)
0°	3.500 - 5.500	3.000 - 5.000
45° Long" Recoil	3.500 - 8.500	3.000 - 80000
45° short Recoil	30500 - 6.500	3.000 - 6.000
Maximum	3.500 - 7.000	3.000 - 6.500

The gymnasticator is set for 15 + 1 second pull back time at 0° elevation and 36.0 ± .2 inches from the battery. This setting is not changed for other elevation simulations-

3.4.1.2 0perating rods. All operating rods shall be inspected for scratches during and after completion of mechanical gymnastication. Continued evidence of rod scratching will result in teardown of the affected cylinder, polishing out the scratch or replacing of the rod, replacing of the seals and cleaning and inspection of the cylinder assembly components. Light scratches that can be polished out and that do not re-occur with continued cycling are not cause for teardown.

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3.4.1.3 pistons, rods and nuts or collars, Pistons, rods, and nuts or collars shall withstand mechanical gymnastication without permanent set or failure of the screw threads.

3.4.2 Functional firing. The Gun Mount shall withstand the functional firing test as specified in 4.6.2 without evidence of malfunction or failure and shall meet the criteria of 3.4.2.1 through 3.4.2.3. During and after functional firing, the gun mount shall meet the requierments of 3.4.5, 3.4.6 and 3.4.7.

3.4.2.1 Counter-recoil (CTRR).

3.4.2.1.1 CTRR velocity. At 1.0 ± .1 inch from the complete in-battery position, the CTRR velocity shall not exceed one and one-haif (l-1/2) feet per second.

3.4.2.1.2 CTRR cycle time. The CTRR cycle time in seconds shall be:

Simulated Elevation	Above 30°F	30°F and Below
0°	.700 - 1.900	.700 - 3.500
Maximum	.900 - 3.900	.900 - 5.900

3.4.2.2 Recoil cycle. The recoil cycle travel (in inches) and total cycle time (in seconds) shall be as follows:

3.4.2.2.1 Recoil travel in inches.

Simulated Elevation	Recoil Travel
0°	27 - 36
Maximum	18 - 27

3.4 .2.2.2 Total cycle time.

Simulated Elevation	Above 30°F	30° and Below
0°	.900 - 2.000	.900 - 3.600
Maximum	1.000 - 4.000	1.000 - 6.000

3.4.2.3 Recoil performance zones. For all functional firing test cycles actual recoil pressure versus time curves shall be within defined recoil performance zones shown in figures 1 and 2.)

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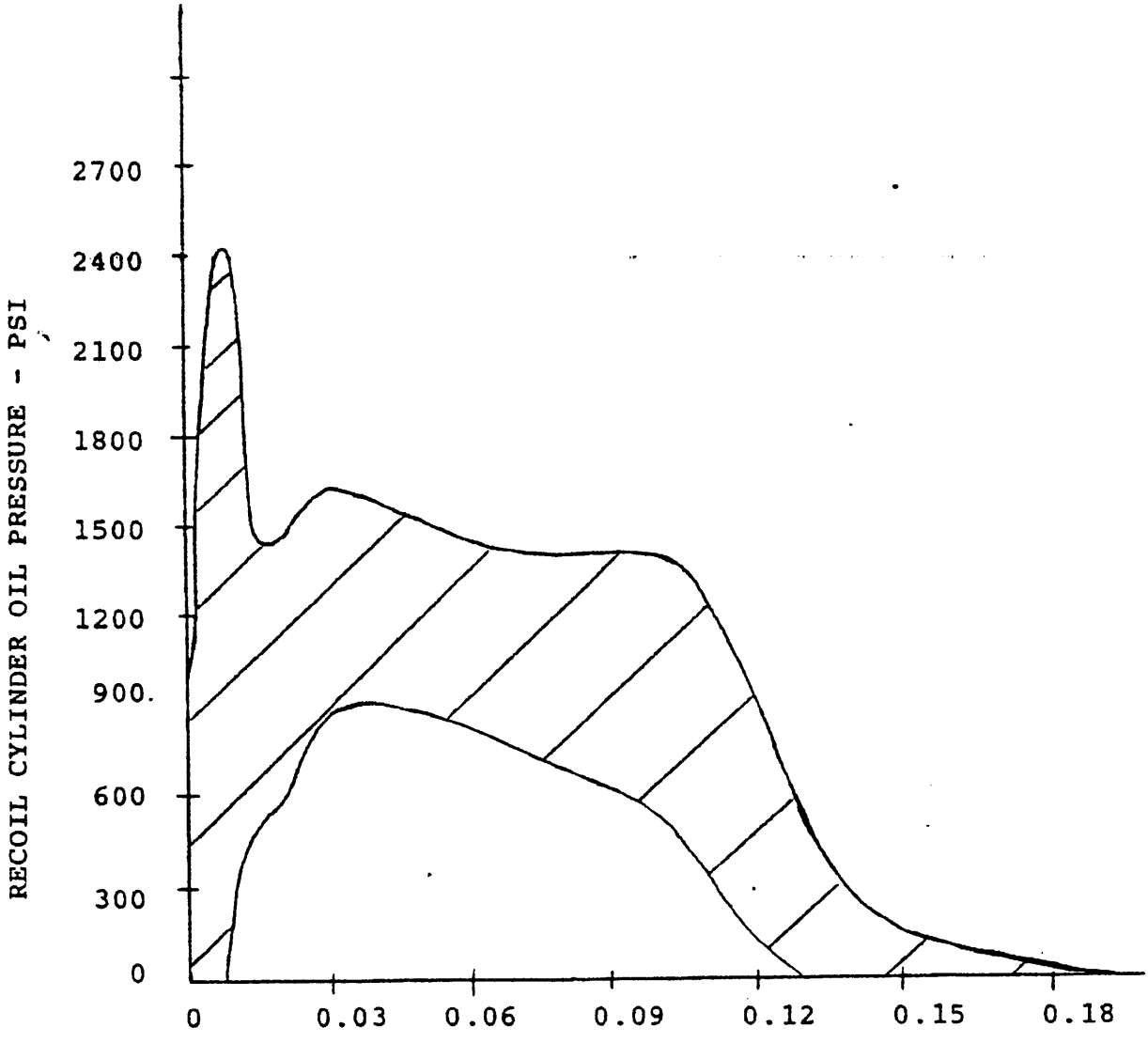


FIGURE 1. Long recoil - functional firing.

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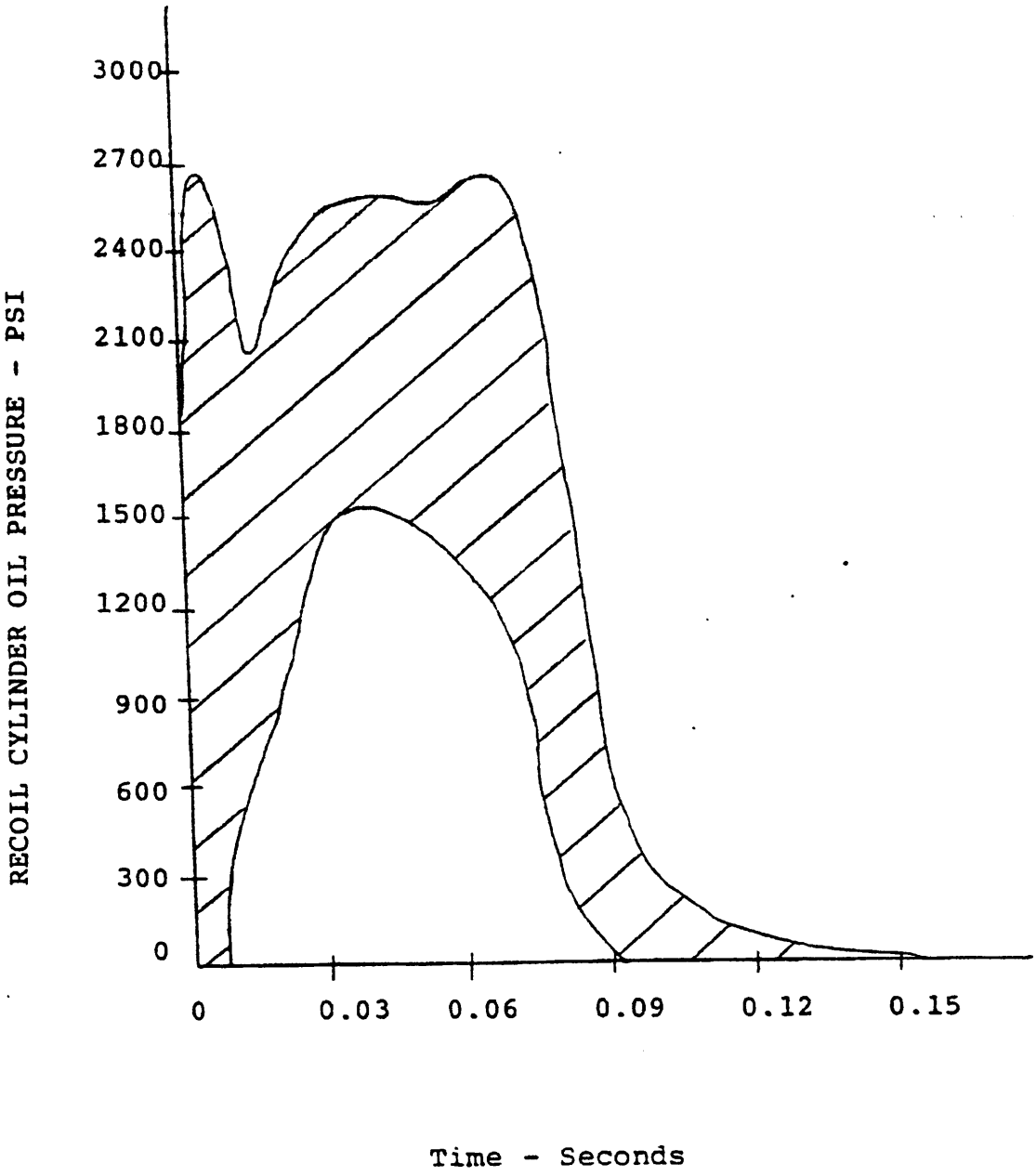


FIGURE 2. Short recoil - functional firing.

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3 .4.3 Proof firing. Each sample mount shall be subjected to proof fixing at a Government proving ground as specified in 4.7.4.2 without evidence of malfunction or failure and shall meet the criteria of 3.4.3.1 through 3.4.3.3. After proof firing, the gun mount shall meet the requirement of 3.4.5, 3.4.6 and 3.4.7.

3.4.3.1 CTRR.

3.4.3.1.1 CTRR velocity. At 1.0 ± .1 inch from the complete in-battery position, the CTRR velocity shall not exceed one and one-half (1-1/2) feet per second.

3.4.3.1.2 CTRR cycle time. The CTRR cycle time at -50°F tp +145°F in seconds shall be:

Elevation in Degrees	CTRR Stroke	
	Above 30° F	30° F and Below
0°	.650 - 1.900	.650 - 3.500
Maximum	.850 - 3.900	.850 - 5.900

3.4.3.2 Recoil cycle. The recoil cyclye travel (in inches) and total cycle time (in seconds) shall be as follows:

3.4.3.2.1 Recoil travel in inches.

Elevation	Recoil Travel
0°	27 - "36
Maximum	18 - 27

3.4.3.2.2 Total cycle time.

Elevation	Total Cycle Time	
	Above 300 F	300 and Below
0°	.800 - 2.100	.800 - 3.700
Maximum	1.000 - 4.100	1.000 - 6.100

NOTE : Conditions of hanging-out-of-battery at the higher elevations are acceptable on the M182 gun mount provided it returns to battery upon lowering tube to loading elevations (approximately 110 roils). The test agency is to record the length of hanging-out-of-battery' s in reports.

3.4.3.3 Recoil performance zones. For all proof firings, actual recoil oil pressure versus time curves shall be within defined recoil performance zones in figures 3 and 4.

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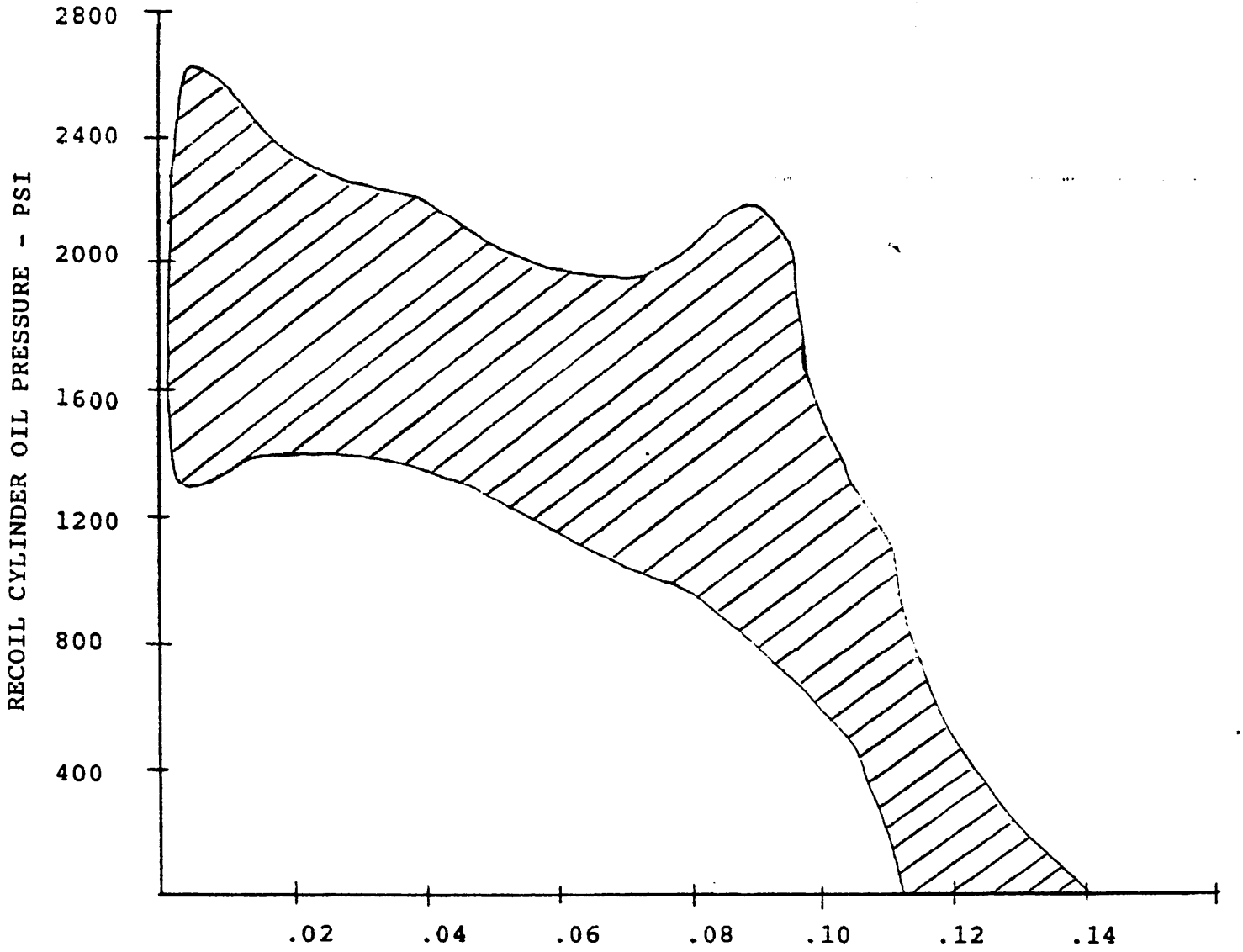


FIGURE 3. Long recoil - proof firing.

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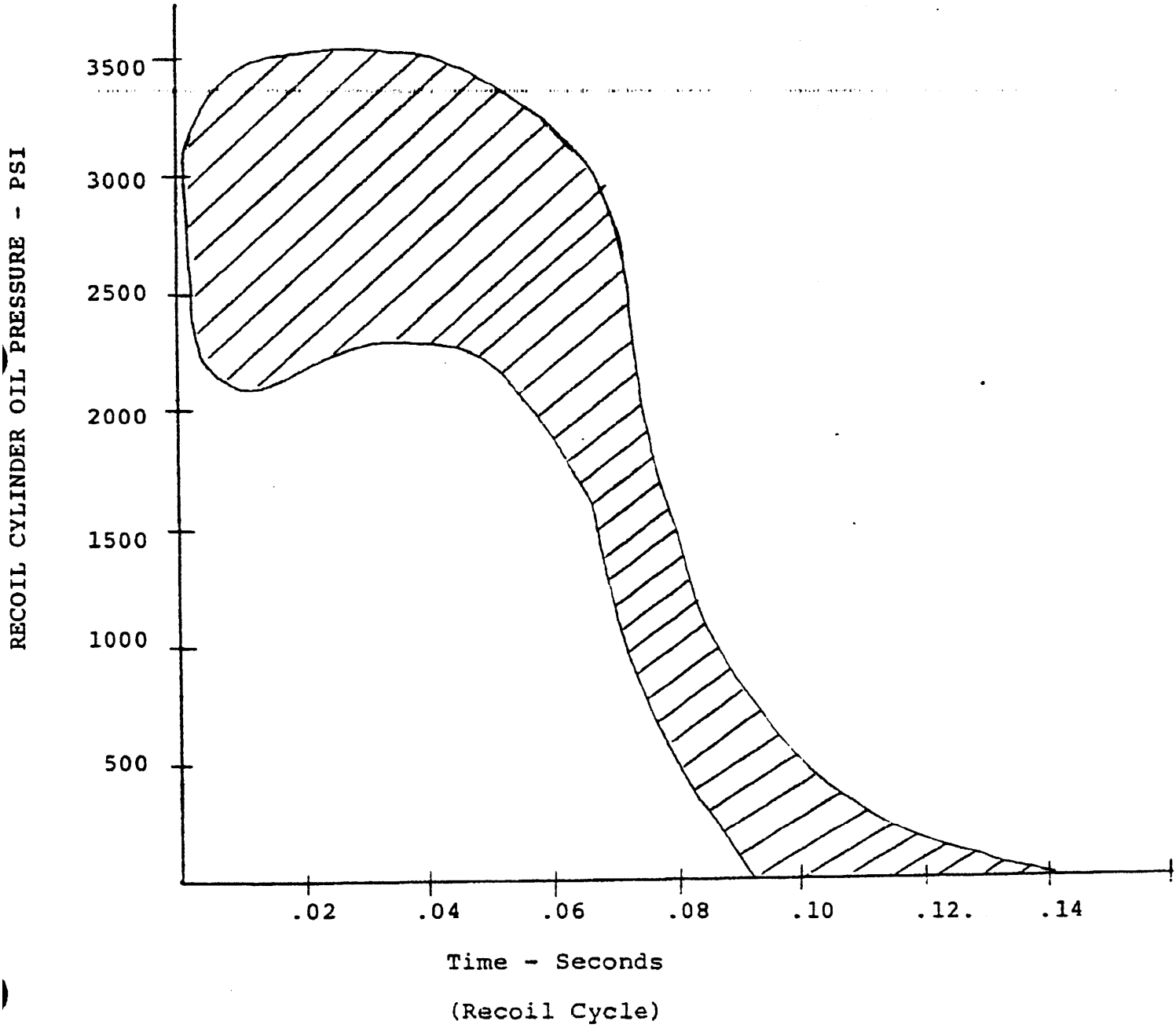


FIGURE 4. Short recoil - proof firing.

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3 .4.4 Climatic conditions. A sample as specified in 4.5.1 shall be proof fired as specified in 3.4.3 at each of the following temperatures: +125°F and -50°F.

3.4.5 Hydraulic leakage.

3.4.5.1 Dynamic. Immediately following cycling of the gun mount there shall be no more than one milliliter (ml) of oil accumulation per every five (5) cycles tested at any seal interface location. Immediately following this determination all oil accumulations are to be wiped clean prior to an additional five (5) cycle incremented test.

3.4.5.2 Static. After the gun mount has set stattically for 24 hours after completion of each test, (mechanical gymnastication, functional firing, proof firing) there shall be no more than 1 ml of oil accumulation at any seal interface location and no evidence of seepage of hydraulic fluid from any of the hydraulic tubes, manifolds, hose connections, and fill/bleeder valves.

3.4.6 Nitrogen leakage. The gun mount recuperator and replenisher shall be examined for leakage of nitrogen gas per 4.5.2.8.

3.4.7 Analysis of data. The data recorded for 3.4.2, functional firing, and for 3.4.3, proof firing, shall be analyzed. If the data recorded for lejngth of recoil and cycle time is not satisfactory, as determined by the procuring agency, the mounts shall be returned to the contractor.

3.5 Protective finishes.

3.5.1 Inorganic. Inorganic finishes, including metal deposited coatings and surface treatments, shall be as specified on the drawings.

3.5.2 Painting. Painting shall be as specified on the drawings.

3.6 Marking. Identification marking, including application of part numbers, and instruction marking shall be as specified on the drawings and in MIL-STD-130.

3.7 Data requirements. The contractor shall generate data in accordance with the requirements of the data item description cited in 4.6.4.

3.8 Workmanship. Each mount shall be free of defects that affect overall appearance, serviceability or performance such as burrs, fins, sharp edges, and surface flaws or blemishes.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for tbe performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Respossibility for compliance. All items must meet all requirements of Section 3 and 5. The inspection set forth in thos specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective makerial.

Quality assurance terms and definitions quality assuracne terms and definitions as used in this specification are in accordance with MIL-STD-109

4.3 Classification of inspection. The inspection requirements specified are classified as follows:

a. First article inspection see (4.4).

b. Quality conformance inspection (see 4.5)

4.4 First article

4.4.1 First article sample. A first article sample shall be the first three manufactured mounts selected and submitted for testing at a location approved by the Government in accordance with the contract requirements (see 6.2). A first article sample shall be selected and submitted for testing at a location approved by the Government in accordance with the contract requirements (see 6.2) when:

a. A new contractor is utilized.

b. A time lapse of one year has occurred since the item was last produced by the contractor.

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c. Significant changes have been made in a manufacturing process or material.

d. A design change is made.

The sample shall be representative of the production processes to be used during quantity production.

4.4.2 First article inspection. The first article sample shall be inspected to determine conformance with the requirements of this specification and any or all requirements of the applicable drawings.

4.4.3 First article rejection. If any sample fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of the sample to comply with the requirements of this specification.

4.5 Quality conformance inspection.

4.5.1 Sampling. Each mount shall be inspected for compliance with the specification except sampling for proof firing as specified in 4.7.4.2 shall be the first three production mounts, and one from each two month's production or one from each twenty-five produced and thereafter. In addition, the first three production mounts shall be climatically fired as specified in 3.4.4 for the first article sample specified in 4.4.1. Failure to comply with the specified requirements shall be cause for rejection.

4.5.2 . Examination.

4.5.2.1 Materials. Materials used in the manufacture of mounts shall be examined and tested in accordance with the material specifications referenced on the applicable drawings. The contractor's inspection records shall be checked to determine conformance to 3.2.

4.5.2.2 Hydraulic system. To determine compliance with 3.3.1, a minimum of 100 ml of oil shall be removed from each recoil cylinder of each mount. The sample shall be taken within five" minutes after gymnastication. Evidence of particles exceeding 3.3.1 shall be cause for rejection. Additionally, failure to comply with the hydraulic leakage requirements of 3.3.1 and 3.4.5 shall be cause for rejection.

4.5.2.3 Surface cleanliness. Each mount shall be examined to determine compliance with the cleanliness requirements specified in 3.3.2. Failure to comply with 3.3.2 shall be cause for rejection of the mount.

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4.5.2.4 Part modification. Each mount shall be examined to determine compliance with requirements specified in 3.3.3. And evidence of parts being altered or hand-fit during assembly beyond the provisions of 3.3.3, shall be cause for rejection of the mount.

4.5.2.5 Air gas and accumulator system. To determine conformance to 3.3.4, the system shall be examined for cleanliness, air and moisture. Failure to comply with 3.3.4 shall be cause for rejection of the mount.

4.5.2.6 Cylinder pressure. To determine compliance with 3.3.5, the cylinders of each mount shall be pressurized in accordance with drawing requirements and inspected for damage, permanent deformation or leakage. Pressures shall be measured with Government-approved gages. Failure to comply with 3.3.5 shall be *cause* for rejection of the mount.

4.5.2.7 Lubrication. Inspection and tests for lubicants shall be in accordance with MIL-G-10924. Failure to comply with MIL-G-10924 shall be cause for rejection of the mount.

4.5.2.8 Gas. Inspection and tests for nitrogen shall be in accordance with BB-N-441. Failure to comply with BB-N-441 shall be cause for rejection of the mount.

4.5.2.9 Visual examination. Each mount shall be examined for the following defects. One or more defects shall be cause for rejection of the mount.

a. Incorrect assembly or missing parts.

b. Discontinuity or absence of protective finish.

c, Incorrect or missing marking.

d. Color of paint not as specified.

e. Workmanship not as specified.

4.5.2.10 Packaging and packing. Examination of packaging and packing shall be in accordance with Packaging Data Sheet P12910860 for M182.

4.6 Testing.

4.6.1 “Mechanical gymnstication. Each mount shall have a mechanical gymnastication in accordance with test method 4.7.3 to determine compliance with requirements specified in 3.4.1.

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4.6.2 Functional firing. Each mount shall be fired in accordance with test method 4.7.4.1 to determine compliance with requirements specified in 3.4.2.

4.6.3 Proof firing. Mounts shall be proof fired as specified in 4.7.4.2 to determine compliance with requirements specified in 3.4.3.

4.6.3.1 In the event a gun mount sample fails to satisfy acceptance criteria the following procedures will be inplemented.

a. The failed sample, accompanied with a detailed report as to the deficiencies found, will be returned **to** the Contractor for immediate corrective action to the represented lot and subsequent production. No dismantling or altering of the mount will occur at the Government test facility without the documented consent of the contractor.

b. The corrected lot shall not be released until such time as three additional mounts selected -by the Government consecutively satisfy acceptance criteria (3.4.3) when tested at the Government proving ground.

c. Three samples from the follow-on production lot will be submitted to the Government proving ground for acceptance " testing.

(1) Acceptance of these three (3) mounts shall permit return to the normal sampling procedure of one mount from each two month's production or one mount from each 25 mounts produced, whichever occurs first.

(2) Rejection **of** any one of the three (3) mounts tested shall require repeating of steps at b and **c.**

4.6.4 Data requirements. The contractor shall submit a quality inspection test, Demonstration and Evaluation report in accordance with 6.2.2 giving the detailed test results for 4.6.1, 4.6.2, and 4.6.3.

4.6.5 Inspection equippment. See MIL_W-63150.

4.7 Methods of inspection.

4.7.1 Test tolerances. All tolerances specified herein are absolute with no allowance made for measuring equipment error (accuracy). Measuremenst shall ensure that the values measured are within the tolerances specified.

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4.7.2 Automatic acceptance inspection. In addition to the requirements as specified. In this specification and in the applicable contract, the requirements of MIL-A-70625 shall apply to automatic acceptance inspection equipment.

4.7.3 Mechanical gymnastication. Each mount, including the cannon or simulated equal shall be gymnasticated on a Government-approved gymnasticator for a minimum of 40 cycles as follows:

a. 10 cycles at 0°, long recoil (36.0 ± .2 inches)

b. 10 cycles at 45°, long recoil (36.0 ± 2 inches)

c. 10 cycles at 45°, short recoil (24.0 ± .2 inchess)

d. 10 cycles at max. QE, short recoil (24.0 ± .2 inches).

Smooth counter-recoil performance is required prior to completion of gymnastication. Recuperator nitrogen charge pressure shall be set at 700 + 10 psi at 70°F temperature prior to gymnastication.

4.7.3.1 Particle count. After gymnastication, oil shall be drained from the assembly for particle count (3.3.1).

4.7.3.2 Hydraulic leakage. During each 10 cycle test the mount shall be lnspected for evid for evidence of hydraulic leakage (3.4.5).

4.7.3.3 Nitrogen leakage. During each 10 cycle test, the mount shall be inspected for evidence of nitrogen leakage (3.4.6).

4.7.4 Functional/proof firing. Each gun mount shall be subjected to the followlng firing test:

4.7.4.1 Functional firing. Each gun mount shall be submitted to the powder gymnastication or hydraulic artillery simulation test as specified in Table II with the recuperator nitrogen pressure set to 700 ± 25 psi at ambient temperature prior to firing. All testing shall be done on a test stand at zero elevation with the variable recoil locked for long recoil or short recoil as applicable to simulate specified elevations. The gymnasticator/simulator shall be varied as required to meet the stated impluse levels. All measurements required by 3.4.2 shall be made for erratic motion in counter-recoil. During and after firing, each mount shall be examined to determine compliance with oil leakage requirements specified in 3.4.5 and nitrogen leakage as specified in 3.4.6.

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TABLE II. Functional firing.

Round Number	Simulated Elevation	Recoil Mode	Impulse Level
1	0°	Long	50%
2	0°	Long	100%
3	Max.	Short	100%
4	Max.	Short	100%
5	0°	Long	100%

NOTE : Impulse applied to the recoil mechanism at 100% impulse level is 6350 pound-second minimum for long recoil and 6800 pound-second minimum for short recoil. At least one cycle in both long and short mode shall meet or exceed the 100 % level. The recoil oil pressure for long recoil and for short recoil should be within the performance zones specified by figures 1 and 2 at the 100% level.

The first three mounts selected as specified in specified in Table III. After the first three mounts, mounts selectd shall be proof fired as specified in Table IV. Recuperator nitrogen pressure shall be set at 700 ± 25 psi at ambient temperature prior to firing. All measurements required by 3.4.3 shall be recorded for each firing. Observation shall be made for erratic motion in counter-recoil. During and after firing, each mount shall be examined to determine compliance with oil leakage requirements specified in 3.4.5 and nitrogen leakage as specified in 3.4.6. .

TABLE III. Proof firing (First article)

Round Number	Elevation	Recoil Mode	Charge
1	0°	Long	M4A2, Zone 6
2	0°	Long	M203., Zone 8
3	Max.	Short	M203, Zone 8
4	Max.	Short	M203, Zone 8
5	0°	Long	M203, Zone 8

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TABLE IV: Proof firing (Lot Acceptance)

Round' Number	Elevation	Recoil Mode	Charge
1	0°	Long	M4A2, Zone 6
2	Max.	Short	M203, Zone 8
3	0°	Long	M203, Zone 8

NOTE : (1) All M4203 charges shall be preconditioned for 24 hours to 145°F prior to Proof firing and a projectile weighing 104 ± 1/2 lb. shall be used. (2) For the first three mounts to be proof fired at + 1250 F and -50° F, fire five rounds at +125° F, and five rounds at -50° F (as specified in Table III).

5. PACKAGING

5.5 Pilot pack. A pilot pack consisting of a complete and packed shipping container, packaged in accordance with Packaging Data Sheet 12576100 to the level of protection specified (see 6.2), shall be forwarded as specified in the contract.

5.2 Preservation , packaging, packing and marking. The mounts shall be unit packaged, packed and marked in accordance with the requirements of Packaging Data Sheet 12910860 ti tehe level of protection specified (see 6.2).

5.3 Repai parts. Repair parts shall be unit packaged, packed and marked in accordance with the requirements of the applicable Packaging Data Sheets (see 6.2).

6. NOTES

(This section contains infOrmation of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These types of hydropneumatic gun mounts are designed for use in combat vehicles.

6.2 ordering data.

6.2.1 Acquisition requirements.

a. Title, number and date of this specification.

b. Title, number and date of applicable documents and drawings.

c. Requirements for submission of first article sample by the contractor (see 3.1 and 4.4).

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d. Place of inspection and test.

e. Levels of packaging required (see 5).

f. Air conditioning for assembly (see 6.3) .

6.2.2 Data requirements. When this specification in sused in an acquisition which incorporate a DD Form 1423, contract Data Requirement List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporate into the contract. When the provisions of DOD Far Supplement Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked, and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data Requirements	Applicable DID
4.6.1, 4.6.2, 4.6.3	Quality Inspection Test Demonstration and Evaluation Rpt.	DI-R-1724

6.3 Assembly areas. Assembly areas, special provisions, and equipment should be provided by the contractor. The assembly area should be air conditioned using an air conditioning unit with a 50 micron filter. The area temperature should be maintained between 68° and 72°F. The air flow rate per cubic inch per minute should be such as to maintain area air pressure at between .75 and 1.5 inches of water. Relative humidity should not exceed 40-50 percent. Areas should be clean and free of dust, lint, and abrasive materials that may become entrapped in the assembled mechanisms. No grinding, filling, scraping, or any operation producing abrasive dust should be performed in the assembly area.

6.4 Functional firing. Information regarding the artillery simulator, see 4.7.4.1.is available from the Commanding Officer, Rock Island Arsenal, Rock Island, IL 61299-6000.

6.5 Hydraulic leakage. The quantity of gun mount hydraulic fluid leakage cannot be precisely measured, therefore the judgement and experience of the inspector will be relied upon for capliance with the criteria of 3.4.5. One milliliter of hydraulic fluid is composed of approximately 25 drops.

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6.5.1 Seal interface locations. The seal interface locations are defined as the area around the following components:

a. Buffer Rod Wiper.

b. Recoil Rod Wiper, (2 places).

c. Variable Recoil Selector Gears, under the Variable Recoil Cover, (2 places).

6.5.2 Hydraulic fluid seepage. Hydraulic fluid seepage is defined as any evidence of hydraulic fluid beyond the seal that does not result in the formation of a droplet.

6.6 Subject term (key word) listing.

Combat vehicle -
Functional firing
Gun mount
Gymnastication
Howitzer
Hydraulic system
Hydropneumatic
Proof firing
Recoil

6.7 Drawings. Drawing listed in Section 2 of this document under the heading US Army Armament Research, Development and Engineering Center (ARDEC0 may also include drawings prepared by, and identified as, Edgewood Arsenal, Frandford Arsenal. Rock Island Arsenal, ARRADCOM or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1025 -A167)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-70821 (AR)	MOUNT, HOWITZER, M182

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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